State of The Art: Fragility Function of Electrical Power System

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***Abstract*— Natural disasters, particularly earthquakes, can cause the electric power system to collapse, which can be brought on by one of the infrastructure breakdowns in the power system. Damaged power system can cause losses to electricity providers, electricity users, so it is necessary to understand the vulnerability of electric power system assets to earthquakes. The research fragility function presents the concept of fragility function in power system of the entire power system, showed fragility of power plant, transmission line, tower transmission, transformers, generator, and substation. Substation is more fragile than the others because substation consist of circuit breaker and disconnecting switch the most fragile asset. In addition, this paper also presents how to creating the fragility function and several solutions to increase the resilience of the electric power system to earthquakes.**

Keywords—Fragility Function, Power system, Earthquake, Damage, Resilience

I. INTRODUCTION

In recent year natural disasters are becoming more frequent, such as earthquakes, tsunamis, floods, hurricanes, windstorm, blizzards, wildfire, volcanic eruptions and more. Earthquake as natural disasters can damage residential buildings, hospitals, bridges, power lines, water facilities, and other important facilities. Earthquakes with large magnitudes produce ground movements that can damage the performance of the electricity power system. Damaged power system can cause losses to electricity providers, electricity users [1] and others. The main Critical Damage by effects of the earthquake at [2] showed that earthquake causes the power disruption and effect to industrial companies and financial losses.

One of the methods used to evaluate and predict the behavior of buildings or infrastructure, including power systems, during earthquakes is performance Based Earthquake Engineering (PBEE). This method focuses on measuring the performance of a structure or system by considering the acceptable level of damage and the functional requirements of the system during and after an earthquake. PBEE involves the analysis and simulation of earthquakes, the use of mathematical models and simulation software to predict system behavior, and the use of fragility curves to relate earthquake intensity to the expected level of damage to the system. We can see PBEE concept in [3], [4], [5], [6], [7], [8], [9].

To determine how much physical damage has occurred at the component or system level in relation to the structural reaction, used fragility function[11] an example of a damage analysis, damage relationship for standard gypsum wallboard partitions[12] or another classification of damage used fragility function in [14], [15], [11], [3], [4], [8], [17],[18].

Using variables like peak acceleration, spectrum acceleration, duration, etc., earthquake engineers describe the intensity of ground vibrations. The ground motion parameters that are most closely related to how the target system responds are the most helpful. It is important to understand that different types of systems have varied optimum parameters for predicting reaction.[8]. Electric power systems, including substations, generation plants, and distribution circuits, are susceptible to damage during earthquakes. The ground shaking caused by seismic events can lead to structural damage and operational disruptions in these critical facilities. The selection of appropriate ground motion parameters for evaluating their vulnerability is essential for assessing seismic risk accurately. Peak Ground Acceleration (PGA) is one of the most crucial ground motion parameters for assessing the seismic vulnerability of electric power system components. It directly correlates with the intensity of ground shaking and is particularly relevant for determining the potential damage to structures and equipment.[19].

The objective of this paper will be shown how the fragility function of the equipment/asset for the electrical power system. This paper is organized as follow: Section II describes the concept of power system. Section III focus on fragility function, commonly used functions of fragility and an example. Section IV explains the function fragility of electrical power system and conclusion in Section V.

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II. POWER SYSTEM

One of a nation's main lifelines that affects how well-functioning key facilities like hospitals, government buildings, and educational institutions run is its electric power infrastructure. Including substations, supervisory control and data collection facilities, and power generation facilities (A power transmission system) that are joined by transmission lines and arranged in a network of transmission systems [20]. It is a crucial infrastructure that enables the production, delivery, and consumption of electricity, which is an essential form of energy for modern society.

A system failure formulation that incorporates the behavioral complexities of distribution and transmission systems will be a valuable tool for future development [21]. According to [19] the power system is defined as circuit consisting of generations, substations, and distribution. When there is an earthquake, all of these parts are susceptible to damage, which could seriously interrupt the power supply.

III. WHAT IS FRAGILITY FUNCTION

Fragility is a fundamental concept used to quantify the probability of failure of components or systems as a function of ground motion intensity, such as Peak Ground Acceleration as failure of nuclear power plant components and the peak ground acceleration due to earthquakes [22], [23]. Fragility functions are valuable tools for understanding the vulnerability of assets to certain types of natural disasters, the unique characteristics of winter storms and other long-duration events require specialized analyses and adaptive approaches to effectively assess and address their impact on the power grid and other critical infrastructure[21].

According to Porter An accurate and comprehensive classification of fragility functions based on the methods used to create them. These classes capture different approaches to developing fragility functions, each with its strengths and limitations, each class in more detail:

1) Empirical fragility functions are created by analysing observational data from laboratory tests or real-world events.
2) Analytical fragility functions are derived by creating and analysing structural models of the asset or class of assets.
3) Expert opinion or judgment-based fragility functions are created by seeking input from experts with experience in the asset class. [23] and,
4) Hybrid fragility functions combine elements from both empirical and analytical methods.[24].

Commonly used fragility functions [23] we can see in equation below. Each of the fragility functions must be developed using the EDP's median and logarithmic standard deviation. [25], and the others function in, [19], [22], [26], [27], [28], [29], [30], [31] :

$$F_d(x) = P[D \geq d | X = x] \qquad d \in \{1, 2, \ldots, N_p\}$$

$$= \Phi\left[\frac{ln\,(x/\theta d}{\beta d}\right] \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Φ represents the cumulative distribution function of the standard normal distribution, $F_d(x)$ a fragility function, X uncertain excitation, D *is* Engineering Demand Parameters (EDPs) are measurable quantities that represent the demands imposed on a structure or system during a hazard event, such as an earthquake or a windstorm. In seismic fragility analysis, EDPs could include peak ground acceleration (PGA) or integratory drift, while in wind fragility analysis, wind speed might be considered as an EDP. d is a particular value of D, nD number of possible damage states, x a particular value of X, βd is the logarithmic standard deviation that characterizes the spread of the fragility function. In the other hand [14] explained to created fragility or we can see at [11][32] [21] using PBEE method [4].

Examples in Fig 1 are seismic fragility curves are graphical depictions that show how likely an asset or system is to fail Probability of Failure (PoF) at various degrees of earthquake-triggered PGA, in Figure 1 we can see disconnecting switch (DSW) 115 KV having a PoF of 20% at 0.5g [33]. We can create an asset fragility function with PGA data but we can also create with high winds, floods, and other natural disasters.

An example of making a fragility curve is taken from data on blackouts in an area of the distribution system in the city of Banda Aceh caused by high winds since January 2019 to Mei 2023. High winds were chosen because it is difficult to obtain earthquake PGA data. This is because earthquake data and damage were not recorded when the earthquake occurred in Banda Aceh. In the example presented, the data taken is conductor fail data that was disturbed by high winds. level winds identified with two is high wind and extreme wind. From damage occurs and how long the repair will take, the median and standard deviation were found using Excel, then the results of the fragility function were obtained which can be seen in figure 2.

In figure 2 we can see the resulting lines were only 2 because the damage was only categorized as slight damage (0,01 – 2.00 hours) and moderate (2,1 – 4,00 hours), we can make categorize until 3 or 4, as long as data complete and available. In this case only 2 damage because for heavy damage no data was found which resulted in the duration of the outage exceeding 4 hours or more.

IV. FRAGILITY FUNCTION OF POWER SYSTEM

Table 1 the research fragility function showed fragility of power plant, transmission line and tower transmission, transformers, generator, substation. Substation is more fragile than the others because substation consist of circuit breaker and disconnecting switch the most fragile asset. Each asset of power system will response earthquake seismic (PGA) and asset will fail based on unique fragility function. Substations, being critical components of the electrical power system, are vulnerable to seismic events, particularly if their components are not adequately anchored or designed to withstand strong ground shaking. Earthquakes can cause significant damages to substations[43].

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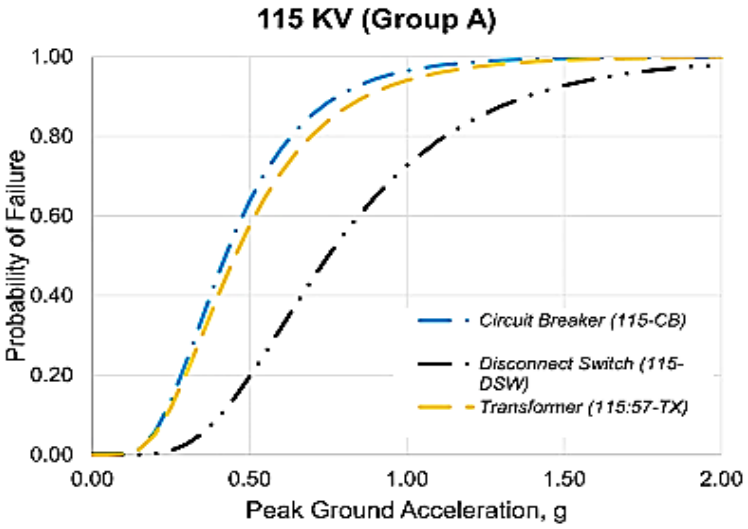


Fig. 1. Blue is fragility curve for circuit breakers, black disconnecting switch and yellow is transformer for 115 kV Voltage level of asset A[33]

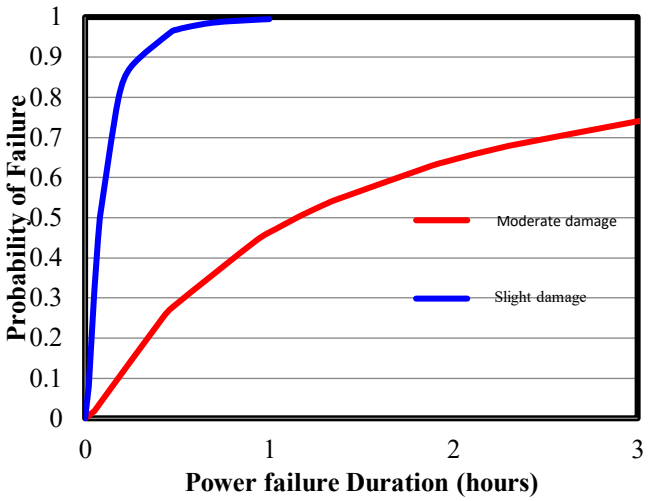


Fig. 2. Fragility curve for conductor in distribution system

Table 1. Fragility function for equipment electrical power system

Year	Component/ Equipment	Fragility Curve Parameter/ Damage Level	Respond	Result
1996[34]	Micro component dan Macro component	Failure Not Failure	PGA	bar-bearing has proven to be the most important micro component (although TAs are more fragile). Assuming lognormal distribution for its fragility, results for the network "as is" are compared with results for the network containing bar bearings (micro component 8) with the mean of the lognormal distribution increased by 30%
1996[26]	Switchyard/ Substation	No Damage Damage	PGA	Most Fragility: Live tank (ATB 7) Circuit Breaker 230 kV (soft soil) Live tank Circuit Breaker 500 kV Live tank (ATB 6) Circuit Breaker, 230 kV (soft soil) Live tank (ATB 6) Circuit Breaker, 230 kV Circuit Breaker (SF 6), 230 kV
2001[21]	Transmission line and Substation	Light: 1 – 10% Moderate: 10 – 30 % Heavy: 30 – 60 % Major to destroyed:60– 100 % replacement value	MMI, PGA and PGV	Transmission tower more fragility than substation.
2012[20]	Substation	Unanchored Substation Low (34.5kV – 150kV) Medium(150kV-350kV) High(350kV-) Unanchored generation Small(<200MW) Medium/Large(>MW)	the fragility curves used for substations (HAZUS ESS2, ESS4, ESS6) and generation facilities (HAZUS EPP2, EPP4)	Substation more fragility than Generator
2020[28]	Power distribution lines	None Slight damage Moderate damage Extensive damage Complete damage	PGA (0.6 g)	0 0.0014 0.413487 0.585113 0.0043 The probability of different damage state if the estimated PGA at the location of the distribution line section is found 0.6g.
2014[35]	Line and Tower transmission	Normal High Extreme	Wind speed	The loss of load frequency (LOLF) in the extreme wind scenario is more than three times greater than the LOLF in the normal wind scenario for the normal network (0.131 and 0.039 occurrences/Sunday).
2012[19]	Substations Distribution Circuits	None (ds1) Slight/minor (ds2) Moderate (ds3) Extensive (ds4) Complete (ds5) None (ds1)	PGA	ds1: no damage ds2: failure 5 % of the disconnect switches ds3: failure 40 % of the disconnect switches, circuit breaker ds4: failure 70 % of the disconnect switches, circuit breaker ds5: failure all of disconnect switches, circuit breaker ds2: failure 4% of all circuits

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Year	Component/ Equipment	Fragility Curve Parameter/ Damage Level	Respond	Result
		Slight/minor (ds2) Moderate (ds3) Extensive (ds4) Complete (ds5)		ds3: failure of 12 % of circuits ds4: failure of 50 % of circuits ds5: failure 80 % of all circuits
	Generation Plants	None (ds1) Slight/minor (ds2) Moderate (ds3) Extensive (ds4) Complete (ds5)		ds2: turbine tripping or light damage to diesel generator ds3: some by the chattering of instrument panel and racks ds4: considerable damage to motor driven pumps ds5: extensive damage to large motor operated valves
2016[36]	Porcelain Bushing Transformer 400 kV	Minor Moderate	PGA	Still in minor and moderate damage, special seismic considerations are required for power transformers in seismic areas (PGAs ≥ 0,5 g)
2019[38]	Power generator	No damage Sligh damage Moderate damage Extensive damage Complete damage	PGA	No damage 100% Sligh damage 100% Moderate damage 50 % Extensive damage 0% Complete damage 0% Power generator zone 3 more fragility then zone 1, 2 and 4.
2020[39]	Substation Distribution Lines	the fragility curves used substations and Distribution lines HAZUS	PGA	Possibility of risk Substation: Cluster 1 (High) Cluster 2 (Low) Possibility of risk Distribution Lines: Cluster 0 (High) Cluster 1 (Nil) Cluster 2 (Medium) Cluster 3 (Low)
2021[40]	Power Transformer 132/20kV Circuit braker 132 kV Current transformer 132 kV Disconnect switch 132 kV Circuit braker 20 kV	Retrofitted Non retrofitted	PGA	The most important components to retrofit are power transformers and the circuit breakers on the distribution (20 kV) feeders.
2020[41]	power line and pipeline	Minor 4%, moderate 12%, extensive 50% and complete 80%	PGA	Optimal hardening plan for protecting the resilience, robustness of transmission lines and loads against seismic hazards. IES is more reliable against seismic attacks than the electricity network.
2019[31]	Braker branches Load Transformers Generators	Damage Not Damage	PGA	the most fragile asset is Breakers and generators are the least
2018[42]	Substation Plant Transmission line	Plants and substations' post-quake conditions are classified as light/moderate, moderate, extensive, and complete.	PGA	This result suggests that the vulnerability of the grid system is more susceptible to substation failure than to the failure of plants and transmission lines.

Pacific earthquake engineering research center (PEER) was compile equipment performance data from past earthquake to analysis of equipment vulnerabilities of substation and compared to current fragility curves developed using expert opinion for establishment of mitigation priorities and result the key parameter to define equipment fragilities from earthquake damage. [3]

FEMA (Federal Emergency Management Agency) categorizes the post-earthquake states of power plants and substations, as well as other critical facilities, based on the extent of damage they experience during seismic events. These categories help in assessing the severity of damage and determining the operational status of the facilities after an earthquake. This level of damage can significantly impact the substation's ability to function as intended[19].

In [44] The design guidelines are used to decrease the power system's vulnerability to earthquakes among other upgrade aging infrastructure and construct to endure greater events than those in the past. Establish a cost- and risk-acceptable level. upgrade vital cables, switchyards, and substations to seismic requirements.

To reduce risks and vulnerabilities related to infrastructure due to earthquakes several engineering solutions have been developed and implemented in Table 2 [45]. Increase infrastructure resilience for earthquake in power system is important thing for resilience of electricity. Methods that help make the power system more resilient, with uses power system planning to harden and improve the design of the infrastructure in order to boost its robustness. Infrastructure in the power system can be hardening with higher design standards for better performance in the event of earthquakes are generally very similar various forms of power infrastructure[43].

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Table 2 Evaluation of the resilience of power system infrastructure to earthquakes

Type	PGA	Asset/equipment	Engineering Improvement	Quality Improvement
Thermal Power Plant (coal, gas, oil)	7 Mw 0,4 g	Item and Their attachments	Seismic component and anchorage	Construction inspection, testing
Hydropower Plants	7 Mw 0.4 g	Intake towers, lift joints, and the gateway	Designing for higher seismic events requires careful consideration of proper anchorage and the use of seismic components	Inspection during construction, periodic inspection
Solar Farm	7 Mw 0.4 g	Support structure	Adequate anchorage, proper design and bracing	Inspection, maintenance
Wind Farm	1 g (large event)	Monopole	Use seismically robust unit	Maintenance, obtain manufacturer test and certificates
Nuclear Power Plants	Large events	Main structures, interior components	Main building seismic insulation, flexible connection Components with seismically rating; pipes, cable shelves, etc.	Testing, inspection, construction documentation
Substations	7 Mw 0.4 g	Bushings, switches, circuit Breakers	Component anchorage, use of seismic components	Review all test documents, ensure redundancy, spares
Transmission and distribution lines	7 Mw 0.4 g	T&D Systems	Use seismic components	Periodic inspection

Electrical network maintenance must be paid attention to in order to increase electrical resilience. Restoring electricity quickly after earthquakes and other natural disasters can be done with power system and infrastructure operations strategies. Backing up electricity supplies can also include making mobile generators, energy storage, distributed resources and microgrids

V. CONCLUSION

This paper presents the concept of fragility function in power system infrastructure due to earthquake natural disasters this concept is used to analyze the vulnerability of power system assets. This paper also shows which assets are most vulnerable to earthquake disasters. An example to create fragility curves in electric power systems are also described here. In addition, developments and solutions to increase earthquake resilience are shown so that power system assets are more earthquake resistant.

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